

05_Hypothesis_Testing

Purpose

The purpose of hypothesis testing is to validate the assumptions defined during the business understanding phase using actual marketplace data. Each hypothesis was evaluated using SQL analysis, KPI calculations, and business performance metrics to determine whether it was supported by the available evidence.

Hypothesis 1

Hypothesis

Certain geographic regions contribute disproportionately to total marketplace revenue.

Evidence

- Top 10 states contributed 87.38% of total marketplace revenue.
- São Paulo (SP), Rio de Janeiro (RJ), and Minas Gerais (MG) generated the highest revenue.

Result

Supported

Conclusion

Marketplace revenue is highly concentrated within a limited number of states. Revenue generation is heavily dependent on a small group of economically significant regions.

Hypothesis 2

Hypothesis

A small number of product categories generate the majority of marketplace sales and revenue.

Evidence

- Bed, Bath & Table, Beauty & Health, Computer Accessories, Furniture & Decoration, and Watches & Gifts were the highest revenue-generating categories.
- The top 10 categories represented 62.96% of the product catalog.

Result

Supported

Conclusion

Marketplace performance is driven by a relatively small group of product categories. These categories contribute a substantial portion of overall sales activity and revenue generation.

Hypothesis 3

Hypothesis

Revenue growth is strongly influenced by repeat customers rather than one-time buyers.

Evidence

- Orders Per Customer: 1.03
- Repeat Purchase Rate: 3.12%

Result

Not Supported

Conclusion

Most customers make only a single purchase. Repeat purchasing behavior is limited, indicating that revenue is primarily generated through customer

acquisition rather than long-term customer retention.

Hypothesis 4

Hypothesis

Operational performance contributes positively to marketplace success.

Evidence

- Delivery Success Rate: 97.02%
- Majority of orders were successfully delivered.

Result

Supported

Conclusion

The marketplace demonstrates strong operational performance and reliable fulfillment processes. Delivery execution is not currently a major business concern.

Hypothesis 5

Hypothesis

Customers are generally satisfied with their purchasing experience.

Evidence

- Average Review Score: 4.09 / 5
- Five-Star Review Rate: 57.78%

Result

Supported

Conclusion

Customer sentiment is generally positive. Most customers report satisfactory experiences and provide favorable ratings following their purchases.

Hypothesis 6

Hypothesis

Seasonal purchasing trends influence monthly revenue performance.

Evidence

- Monthly revenue varied significantly across the analysis period.
- Revenue increased substantially during several periods in 2017 and 2018.
- March 2018 recorded the highest monthly revenue.

Result

Supported

Conclusion

Customer purchasing activity varies across different periods of the year, indicating the presence of seasonal demand patterns that influence marketplace revenue performance.

Overall Assessment

Six key business hypotheses were tested using SQL analysis and KPI evaluation.

Results showed that marketplace performance is heavily influenced by geographic concentration, category concentration, operational efficiency, customer satisfaction, and seasonal purchasing behavior.

The only unsupported hypothesis was customer retention. While customer acquisition performance is strong, repeat purchasing behavior remains low, making customer retention the most significant improvement opportunity identified during the analysis.

Overall, the analysis confirms that the marketplace operates efficiently, generates strong customer satisfaction, and achieves substantial revenue

performance, while also highlighting customer retention as a critical strategic focus area for future growth.